

Charter/Spectrum

Wall-Mount Condenser Coil Cleaning Standard

Please take a few minutes to watch the two attached videos. They demonstrate the proper way to clean the condenser coils on the Bard-style wall-mount units we are seeing at many Charter/Spectrum sites. Many of these small, remote server buildings do not have a water supply, so we need a consistent process that cleans the coil thoroughly without damaging it.

Key Points From the Videos

1. Clean the coil from the INSIDE OUT.

This is probably the most important takeaway. The technician does wash both sides of the coil, but he specifically emphasizes cleaning from the inside out. That allows us to push dirt and debris back out of the coil instead of driving it deeper into it.

2. Do NOT use a pressure washer.

These coils have thin aluminum fins that are very easy to bend. We have already seen fins damaged from excessive pressure. A pressure washer may provide plenty of water, but the pressure can do more harm than good. Most traditional coil-washing trailers use pressure washers, which is one reason they are not ideal for these particular units.

New Water Setup We Are Working On

We are currently working on a setup specifically for these sites that will:

- Hold approximately 40 gallons of water
- Mount directly into the truck's trailer hitch
- Supply approximately 60 PSI
- Use a wand with a 90-degree spray tip

The 90-degree tip is important because spraying through the opening at an angle increases the chance of folding fins over and also reduces how deeply the water penetrates the coil.

Fan Shroud / Housing Access

Notice in the cleaning video that the technician completely removes the fan shroud assembly. I have done this before, and it is not always as easy as the video makes it look. If we can safely remove the entire assembly, that will give us the best possible access and the lowest chance of damaging the coil.

However, the fan housing will also slide back fairly easily from the non-compressor side of the unit. With the proper 90-degree wand, this should give us plenty of access to clean the coil correctly without completely removing the housing.

If you are at a site without one of our water-tank setups, removing the entire housing may still be the best option for getting proper access.

Coil Cleaner - Pump Sprayer Required

Use a strong, heavy-foaming condenser coil cleaner such as Nu-Brite or Kleen-Foam. Apply it with a **pump sprayer - NOT a foaming coil gun**.

Why: The cleaner needs to penetrate deep into the coil before it starts foaming. A foaming coil gun activates the foaming reaction too early, before the cleaner can work its way into the depth of the coil.

- Nu-Brite - the purple/blue cleaner
- Kleen-Foam - aka the 'brown frown'... because it's brown and you'll definitely be frowning if you get it on you!

PPE REQUIRED: These are strong, corrosive chemicals. Wear the proper PPE and follow the chemical manufacturer's directions whenever handling or applying them.

Verify Airflow After Cleaning

After the coil has been cleaned and the unit is operating, use a **good air temperature probe** to check the condenser air inlet and outlet temperatures. This gives us a simple way to confirm that airflow through the coil has improved and is acceptable.

- The condenser air inlet is typically through the sides of these units.
- Unlike many condensers, the fan on these units **pushes** air through the condenser coil instead of pulling air through it.
- Compare the inlet temperature to the outlet temperature. The temperature difference should ideally be **30 degrees F or less**.
- The fan is designed to push more air through the coil around the **edges** than directly through the center, so probe placement matters.
- **Please follow the videos for proper probe placement.**

For Anyone Running Charter/Spectrum Calls

Please make sure you have:

- A heavy-foaming condenser coil cleaner such as Nu-Brite or Kleen-Foam
- **A pump sprayer for applying the cleaner - do not use a foaming coil gun**
- **A good air temperature probe** for inlet/outlet temperature verification
- Proper PPE
- The proper water-cleaning setup when available

The goal is not simply to make the outside of the condenser look clean. We need to actually clean through the depth of the coil without damaging the aluminum fins, then verify that the coil has proper airflow. These sites rely heavily on these wall-mount units to keep their server/equipment rooms cool, so proper condenser cleaning is extremely important.

Please watch both attached videos and use this process as our standard going forward on Charter/Spectrum locations.

Scan to watch the videos

Use your phone camera

Video 1



First video

Video 2



Second video